# Execution Output – Model Training and Evaluation

```
PS C:\Users\pooja\OneDrive\Desktop\Gauge_Project> python -u "c:\Users\pooja\OneDrive\Desktop\Gauge_Project\train.py"
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1788784673.853783   23052 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation o
rders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1788784677.381752   23052 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation o
rders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
JSON Path: C:\Users\pooja\OneDrive\Desktop\Gauge_Project\dataset\labels.json
Total Records: 290
First Record: {'image': '0-100/1.png', 'gauge_type': '0-100', 'min_pressure': 0, 'max_pressure': 100, 'needle_angle': 150, 'pressure': '0-20', 'status': 'Unsafe(Low Pressure)'}

Encoded Labels:
[ 0  0  0  0  5  5  7  9  9  9  9 10 10  0  0  5  5  5  5  7]

Pressure Classes:
['0-20' '0-50' '101-150' '151-200' '201-250' '21-40' '251-300' '41-60'
 '51-100' '61-80' '81-100']

Label Counts:
Counter({np.int64(1): 45, np.int64(10): 43, np.int64(0): 31, np.int64(6): 30, np.int64(5): 26, np.int64(9): 26, np.int64(8): 25, np.int64(4): 25, np.int64(7): 19, np.int64(2): 11, np.int64(3): 9})
C:\Users\pooja\AppData\Local\Programs\Python\Python312\Lib\site-packages\keras\src\layers\convolutional\base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. Whe
n using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
I0000 00:00:1788784687.147645   23052 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML
plugin.
Model: "sequential"
```

**Model: "sequential"**

| Layer (type) | Output Shape | Param # |
|---|---|---|
| conv2d (Conv2D) | (None, 94, 94, 32) | 896 |
| max_pooling2d (MaxPooling2D) | (None, 47, 47, 32) | 0 |
| conv2d_1 (Conv2D) | (None, 45, 45, 64) | 18,496 |
| max_pooling2d_1 (MaxPooling2D) | (None, 22, 22, 64) | 0 |
| flatten (Flatten) | (None, 30976) | 0 |
| dense (Dense) | (None, 128) | 3,965,056 |
| dense_1 (Dense) | (None, 11) | 1,419 |

```
 Total params: 3,985,867 (15.20 MB)
 Trainable params: 3,985,867 (15.20 MB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 3s 82ms/step - accuracy: 0.1767 - loss: 2.4770 - val_accuracy: 0.2069 - val_loss: 2.2010
Epoch 2/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 72ms/step - accuracy: 0.3017 - loss: 2.1140 - val_accuracy: 0.4138 - val_loss: 1.9429
Epoch 3/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 75ms/step - accuracy: 0.5259 - loss: 1.5651 - val_accuracy: 0.5172 - val_loss: 1.6718
Epoch 4/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 69ms/step - accuracy: 0.6940 - loss: 0.9754 - val_accuracy: 0.5172 - val_loss: 1.5386
Epoch 5/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 71ms/step - accuracy: 0.8405 - loss: 0.6150 - val_accuracy: 0.5862 - val_loss: 1.4743
Epoch 6/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 73ms/step - accuracy: 0.8966 - loss: 0.4446 - val_accuracy: 0.5862 - val_loss: 1.5444
Epoch 7/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 72ms/step - accuracy: 0.9095 - loss: 0.3165 - val_accuracy: 0.6034 - val_loss: 1.5857
Epoch 8/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 71ms/step - accuracy: 0.9440 - loss: 0.2306 - val_accuracy: 0.6207 - val_loss: 1.9572
Epoch 9/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 75ms/step - accuracy: 0.9181 - loss: 0.1915 - val_accuracy: 0.6034 - val_loss: 1.7312
Epoch 10/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 74ms/step - accuracy: 0.9353 - loss: 0.1793 - val_accuracy: 0.6034 - val_loss: 1.7997
Epoch 11/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 75ms/step - accuracy: 0.9181 - loss: 0.2196 - val_accuracy: 0.6207 - val_loss: 1.8303
Epoch 12/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 73ms/step - accuracy: 0.9353 - loss: 0.1723 - val_accuracy: 0.5862 - val_loss: 1.7600
Epoch 13/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 78ms/step - accuracy: 0.9310 - loss: 0.1677 - val_accuracy: 0.6207 - val_loss: 1.9054
Epoch 14/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 72ms/step - accuracy: 0.9353 - loss: 0.1620 - val_accuracy: 0.6034 - val_loss: 1.7900
Epoch 15/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 69ms/step - accuracy: 0.9353 - loss: 0.1562 - val_accuracy: 0.6034 - val_loss: 2.0473
Epoch 16/20
```

```
Epoch 16/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 60ms/step - accuracy: 0.9181 - loss: 0.1505 - val_accuracy: 0.6034 - val_loss: 1.7451
Epoch 17/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - accuracy: 0.9267 - loss: 0.1368 - val_accuracy: 0.6034 - val_loss: 1.9917
Epoch 18/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - accuracy: 0.9138 - loss: 0.1455 - val_accuracy: 0.6207 - val_loss: 1.8545
Epoch 19/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 60ms/step - accuracy: 0.9267 - loss: 0.1418 - val_accuracy: 0.6034 - val_loss: 1.9959
Epoch 20/20
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 59ms/step - accuracy: 0.9310 - loss: 0.1445 - val_accuracy: 0.6034 - val_loss: 1.9354
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - accuracy: 0.6034 - loss: 1.9354

========= MODEL EVALUATION =========
Test Accuracy : 0.6034482717514038
Test Loss     : 1.9354180097579956
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step
```

```
========== DETAILED EVALUATION ==========

CORRECT
Image              : 0-100/85.png
Actual Pressure    : 81-100
Predicted Pressure : 81-100
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (High Pressure)
----------------------------------------
CORRECT
Image              : 0-300/114.png
Actual Pressure    : 0-50
Predicted Pressure : 0-50
Actual Status      : UNSAFE (Low Pressure)
Predicted Status   : UNSAFE (Low Pressure)
----------------------------------------
CORRECT
Image              : 0-100/46.png
Actual Pressure    : 81-100
Predicted Pressure : 81-100
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (High Pressure)
----------------------------------------
WRONG
Image              : 0-300/31.png
Actual Pressure    : 251-300
Predicted Pressure : 0-50
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (Low Pressure)
```

```
----------------------------------------
WRONG
Image              : 0-100/144.png
Actual Pressure    : 81-100
Predicted Pressure : 0-20
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (Low Pressure)
----------------------------------------
WRONG
Image              : 0-100/126.png
Actual Pressure    : 81-100
Predicted Pressure : 61-80
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : SAFE
----------------------------------------
WRONG
Image              : 0-300/80.png
Actual Pressure    : 201-250
Predicted Pressure : 0-50
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (Low Pressure)
----------------------------------------
WRONG
Image              : 0-100/119.png
Actual Pressure    : 21-40
Predicted Pressure : 41-60
Actual Status      : SAFE
Predicted Status   : SAFE
----------------------------------------
CORRECT
Image              : 0-100/10.png
Actual Pressure    : 61-80
Predicted Pressure : 61-80
Actual Status      : SAFE
Predicted Status   : SAFE
```

```
--------------------------------------
WRONG
Image              : 0-300/74.png
Actual Pressure    : 51-100
Predicted Pressure : 151-200
Actual Status      : SAFE
Predicted Status   : SAFE
--------------------------------------
CORRECT
Image              : 0-100/78.png
Actual Pressure    : 61-80
Predicted Pressure : 61-80
Actual Status      : SAFE
Predicted Status   : SAFE
--------------------------------------
CORRECT
Image              : 0-100/47.png
Actual Pressure    : 81-100
Predicted Pressure : 81-100
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (High Pressure)
--------------------------------------
CORRECT
Image              : 0-100/6.png
Actual Pressure    : 21-40
Predicted Pressure : 21-40
Actual Status      : SAFE
Predicted Status   : SAFE
--------------------------------------
WRONG
Image              : 0-300/84.png
Actual Pressure    : 151-200
Predicted Pressure : 0-50
Actual Status      : SAFE
Predicted Status   : UNSAFE (Low Pressure)
```

```
--------------------------------------
CORRECT
Image              : 0-100/64.png
Actual Pressure    : 0-20
Predicted Pressure : 0-20
Actual Status      : UNSAFE (Low Pressure)
Predicted Status   : UNSAFE (Low Pressure)
--------------------------------------
CORRECT
Image              : 0-300/2.png
Actual Pressure    : 0-50
Predicted Pressure : 0-50
Actual Status      : UNSAFE (Low Pressure)
Predicted Status   : UNSAFE (Low Pressure)
--------------------------------------
CORRECT
Image              : 0-100/67.png
Actual Pressure    : 21-40
Predicted Pressure : 21-40
Actual Status      : SAFE
Predicted Status   : SAFE
--------------------------------------
CORRECT
Image              : 0-300/123.png
Actual Pressure    : 201-250
Predicted Pressure : 201-250
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (High Pressure)
--------------------------------------
CORRECT
Image              : 0-300/14.png
Actual Pressure    : 251-300
Predicted Pressure : 251-300
Actual Status      : UNSAFE (High Pressure)
Predicted Status   : UNSAFE (High Pressure)
```

```
----------------------------------------

========== FINAL SUMMARY ==========
Total Test Images : 58
Correct Images    : 35
Wrong Images      : 23
Accuracy          : 60.34%

Model saved successfully!

========== DATASET INFORMATION ==========
Images Loaded : 290
Labels Loaded : 290
Image Shape   : (290, 96, 96, 3)
Label Shape   : (290,)

========== DATASET SPLIT ==========
Training Images : (232, 96, 96, 3)
Testing Images  : (58, 96, 96, 3)
Training Labels : (232,)
Testing Labels  : (58,)
========================================
PS C:\Users\pooja\OneDrive\Desktop\Gauge_Project>
```

```
  warnings.warn(
W0000 00:00:1788786883.816228    4304 tf_tfl_flatbuffer_helpers.cc:364] Ignored output_format.
W0000 00:00:1788786883.816577    4304 tf_tfl_flatbuffer_helpers.cc:367] Ignored drop_control_dependency.
I0000 00:00:1788786883.817834    4304 reader.cc:83] Reading SavedModel from: C:\Users\pooja\AppData\Local\Temp\tmp9jlzfujv
I0000 00:00:1788786883.819143    4304 reader.cc:52] Reading meta graph with tags { serve }
I0000 00:00:1788786883.819320    4304 reader.cc:147] Reading SavedModel debug info (if present) from: C:\Users\pooja\AppData\Local\Temp\tmp9jlzfujv
I0000 00:00:1788786883.823990    4304 mlir_graph_optimization_pass.cc:437] MLIR V1 optimization pass is not enabled
I0000 00:00:1788786883.825085    4304 loader.cc:236] Restoring SavedModel bundle.
I0000 00:00:1788786883.866005    4304 loader.cc:220] Running initialization op on SavedModel bundle at path: C:\Users\pooja\AppData\Local\Temp\tmp9jlzfujv
I0000 00:00:1788786883.873586    4304 loader.cc:471] SavedModel load for tags { serve }; Status: success: OK. Took 55775 microseconds.
I0000 00:00:1788786883.890155    4304 dump_mlir_util.cc:269] disabling MLIR crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, output_inference_type: INT8
W0000 00:00:1788786884.261843    4304 flatbuffer_export.cc:3851] Skipping runtime version metadata in the model. This will be generated by the exporter.

Saved: model_quantized.tflite
TFLite size: 23984 bytes
TFLite size: 23.421875 KB

========================================
          CREATING model.h
========================================
Saved: model.h

========================================
              COMPLETE
========================================

Final files:
model_small.keras
model_quantized.tflite
model.h
labels.txt

Model size: 23.421875 KB
PS C:\Users\pooja\OneDrive\Desktop\Gauge_Project>
```

## TensorFlow Lite Model Conversion

```
I0000 00:00:1788786242.512057   21432 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation o
rders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1788786244.628438   21432 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML
plugin.
Saved artifact at 'C:\Users\pooja\AppData\Local\Temp\tmpaapqph7x'. The following endpoints are available:

* Endpoint 'serve'
  args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 96, 96, 3), dtype=tf.float32, name='input_layer')
Output Type:
  TensorSpec(shape=(None, 11), dtype=tf.float32, name=None)
Captures:
  2262451046352: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262451045776: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480454288: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480455056: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480457552: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480458320: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480454096: TensorSpec(shape=(), dtype=tf.resource, name=None)
  2262480457168: TensorSpec(shape=(), dtype=tf.resource, name=None)
W0000 00:00:1788786245.427492   21432 tf_tfl_flatbuffer_helpers.cc:364] Ignored output_format.
W0000 00:00:1788786245.427619   21432 tf_tfl_flatbuffer_helpers.cc:367] Ignored drop_control_dependency.
I0000 00:00:1788786245.428854   21432 reader.cc:83] Reading SavedModel from: C:\Users\pooja\AppData\Local\Temp\tmpaapqph7x
I0000 00:00:1788786245.429599   21432 reader.cc:52] Reading meta graph with tags { serve }
I0000 00:00:1788786245.429673   21432 reader.cc:147] Reading SavedModel debug info (if present) from: C:\Users\pooja\AppData\Local\Temp\tmpaapqph7x
I0000 00:00:1788786245.434668   21432 mlir_graph_optimization_pass.cc:437] MLIR V1 optimization pass is not enabled
I0000 00:00:1788786245.435329   21432 loader.cc:236] Restoring SavedModel bundle.
I0000 00:00:1788786245.487186   21432 loader.cc:220] Running initialization op on SavedModel bundle at path: C:\Users\pooja\AppData\Local\Temp\tmpaapqph7x
I0000 00:00:1788786245.493635   21432 loader.cc:471] SavedModel load for tags { serve }; Status: success: OK. Took 65380 microseconds.
I0000 00:00:1788786245.543416   21432 dump_mlir_util.cc:269] disabling MLIR crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
TFLite model created successfully!
Model size: 15947856 bytes
PS C:\Users\pooja\OneDrive\Desktop\Gauge_Project>
```

## Arduino Flash Upload Progress

```
Sketch uses 177688 bytes (18%) of program storage space. Maximum is 983040 bytes.
Global variables use 148944 bytes (56%) of dynamic memory, leaving 113200 bytes for local variables. Maximum is 262144 bytes.
Device       : nRF52840-QIAA
Version      : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]
Address      : 0x0
Pages        : 256
Page Size    : 4096 bytes
Total Size   : 1024KB
Planes       : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.000 seconds
Write 177696 bytes to flash (44 pages)

[                              ] 0% (0/44 pages)
[=                             ] 4% (2/44 pages)
[==                            ] 6% (3/44 pages)
[==                            ] 9% (4/44 pages)
[===                           ] 11% (5/44 pages)
[====                          ] 13% (6/44 pages)
[====                          ] 15% (7/44 pages)
[=====                         ] 18% (8/44 pages)
[======                        ] 20% (9/44 pages)
[======                        ] 22% (10/44 pages)
[=======                       ] 25% (11/44 pages)
[========                      ] 27% (12/44 pages)
[========                      ] 29% (13/44 pages)
[=========                     ] 31% (14/44 pages)
[==========                    ] 34% (15/44 pages)
```

```
[==========                    ] 34% (15/44 pages)
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[============                  ] 40% (18/44 pages)
[============                  ] 43% (19/44 pages)
[=============                 ] 45% (20/44 pages)
[==============                ] 47% (21/44 pages)
[===============               ] 50% (22/44 pages)
[===============               ] 52% (23/44 pages)
[================              ] 54% (24/44 pages)
[=================             ] 56% (25/44 pages)
[=================             ] 59% (26/44 pages)
[==================            ] 61% (27/44 pages)
[===================           ] 63% (28/44 pages)
[===================           ] 65% (29/44 pages)
[====================          ] 68% (30/44 pages)
[=====================         ] 70% (31/44 pages)
[=====================         ] 72% (32/44 pages)
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[========================      ] 81% (36/44 pages)
[=========================     ] 84% (37/44 pages)
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[==========================    ] 88% (39/44 pages)
[===========================   ] 90% (40/44 pages)
[===========================   ] 93% (41/44 pages)
[============================  ] 95% (42/44 pages)
[============================= ] 97% (43/44 pages)
[==============================] 100% (44/44 pages)
Done in 7.434 seconds
```

## TensorFlow Lite Micro Tensor Allocation Issue

```
=================================
 Gauge TinyML Deployment
 Arduino Nano 33 BLE Sense
=================================
Loading model...
Model loaded successfully.
Allocating tensors...
```

The trained model was successfully loaded on the Arduino Nano 33 BLE Sense, but the program stopped during tensor allocation. This prevented the model from proceeding to the inference stage.